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Attitude Dynamics and Control of a Spacecraft like a Robotic Manipulator when implementing on-orbit servicing.

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Abstract

The new aspect of the attitude dynamics and control for robot manipulators while performing on-orbit servicing is that they operate on non fixed bases. Even when the robotic manipulator is mounted on a spacecraft, such spacecraft is not fixed in space. The working space is characterized by the microgravity environment. In this environment the spacecraft fluctuates and its rotational motion may be excited by any internal and external disturbances. The complete system, i.e., the spacecraft and the associated robotic manipulator, fluctuate and is sensible to any reaction force and torque inherent to the EVA (extravehicular activities). In this sense all effort done by the robot may result in torque about its center of mass and even result in forces that can cause translational motion of robotic manipulator center of mass. This paper focuses on such scenario by analyzing the impact of the robot arms dynamics on the attitude motion and the associated control effort to keep the attitude motion stable during the manipulator operation. The focus of the dynamics analysis is the close proximity. In such configuration of chaser-target spacecrafts the linear system equations for the translational relative motion is appropriated for the dynamics analysis. The computer simulations are implemented for the relative translational and rotational (attitude motion). The relative attitude equations representing the spacecraft like a robotic manipulator have been simulated through computer by using the MatLab software package. The PID control technique is used to keep the attitude motion stabilized while the robot arms moves its arms under the effect of gravity-gradient and applies forces to execute some hypothetical tasks. The control effort is analyzed to bring about the necessary actuators to accomplish the spacecraft attitude stabilization while providing on-orbit servicing.

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Introduction

The first rendezvous and docking between two spacecraft took place on March 16, 1966. At that time Neil Armstrong and Dave Scott performed a rendezvous in a Gemini vehicle, manually. Since then the operation has been used in in several space missions. Presently RVD/B is frequently used for OOS in the International Space Station, ISS.

Technology for Rendezvous and docking or berthing (RVD/B) is classified as safety-critical. The technology includes high precision sensors and actuators. The great risk is a collision between the chaser and the target spacecraft. There is a critical relative distance between the chaser and the target spacecraft for which there is no possibility to abort the operation and the success will then depends on a careful planning and execution of the operation. On the other RVD/B operation is becoming more and more required in the scenario of the space exploration that nowadays in not restrict to the Earth orbit anymore. Missions are presently extended to other planets and asteroids/comets as well. Future missions are including re-docking to get back to the Earth coming from distant planets or other bodies in the solar system. To be short the RVD/B dynamics analysis and technology are key issues in on-orbit servicing (OOS) missions such those aiming assembling of large space structures, re-supplying of orbital platforms and space stations, exchanging of crews in orbital stations, repairing of orbiting spacecraft, retrieval of spacecraft, re-docking of orbiting vehicle with a lander module when it is returning from a mission on ground of celestial bodies such as Moon and Mars, for example.

The RVD/B process [*1*] involves a series of orbital maneuvers and controlled trajectories for bringing the chaser vehicle into the vicinity of the target vehicle and, eventually to complete the docking or berthing of both vehicles. The last part of the approaching trajectory leads the chaser into narrow boundaries for position, velocity, and attitude and associated rates so as to satisfy the mating process. To clarify the concepts consider the following:

- For docking the chaser Guidance Navigation and Control (GNS) subsystem controls the vehicle state parameters required for entry into the docking interfaces (docking port) of the target vehicle aiming the mating of both spacecraft
- Berthing refers to the operation of parking the chaser at a specific point that may be very close to the target spacecraft. Parking is the scenario of zero relative velocity between both spacecraft. The berthing may or not be followed by a docking.

One important aspect of the dynamics in the vicinity of the target is to consider the relative motion between the vehicles. The relative motion concept allows the parking of the chaser very close to the target without risk of collision. Everything happen as if both spacecraft were stopped with respect to each other in spite of the fact that both of them are movement around the Earth with very high speed. Regarding the attitude motion it is necessary to consider the relative attitude motion if the chaser is to grasp or to lock to the target by using its docking port. If the relative motion is not synchronized the chaser cannot mate the target successfully.

For the sake of dynamics analysis the attitude equations must be written in terms of the relative rotation between both spacecraft. In this paper the equation of relative motion for both the translational and rotational motion is considered. In implementing this study we have considered that initial scenario is that of the close vicinity. The orbital motion presented in this paper is the linear expressions for the relative motion the dynamics analysis aims the scenario of relative velocity zero and relative position about a half meter distant from the target.

In some cases the target spacecraft may impose safety zones, approaching-trajectory corridors and restrict points along the final trajectory to check out vehicles functions and other conditions. For safeness reasons the ground station may assume the process and control the final approach. It is necessary to consider if the host spacecraft is a cooperative target or not. If it is cooperative then the chaser may have to wait for the crew of the target to proceed the approximation and the final mating.

During the operation of OOS the operation of the robotic manipulator disturbs the satellite's platform on which it is mounted. The perturbation acts on both the translation and rotation motion. The spacecraft is excited by the manipulator motion due to absence of friction and the orbital microgravity environment. Everything floats and there is an apparent lack of weight. Because of this the vehicle platform reacts to the torques and force impressed on it by the robot manipulator.

This work deals with orbit and attitude control of a chaser spacecraft when approaching in the vicinity of the target spacecraft. In this study the chaser spacecraft is conceived as a spacecraft like robotic manipulator. For the relative translational motion the equations are assumed to be linear. The idea is to transfer the chaser to the vicinity of the target safely by controlling the relative translational motion. To prepare the spacecraft for docking/berthing the attitude must also be controlled so as to synchronize the attitude of both spacecraft. So the rotational equations of motion are written in terms of the relative attitude. Once the chaser spacecraft is safely berthed by side of the target, the robot manipulator will start moving and attitude control subsystem must keep the platform stable during the manipulator operations. The control effort to keep the chaser platform stable when the robotic manipulator is out of service is compared with the effort to keep it stable when the robotic manipulator is working. Also we have considered a particular case in which the robot links deploy with the same angle but in reverse direction. The idea is that the reverse motion nulls the reactions on the chaser platform, alleviating the control effort.

The details of the translational and rotational motion are presented in the next subsection

Mathematical Modeling

Two set of equations of motion are derived, one for the translational and the other the rotational motion. No coupling between the motions. Figure *1* and *2* shows the configuration assumed for both set of equations.

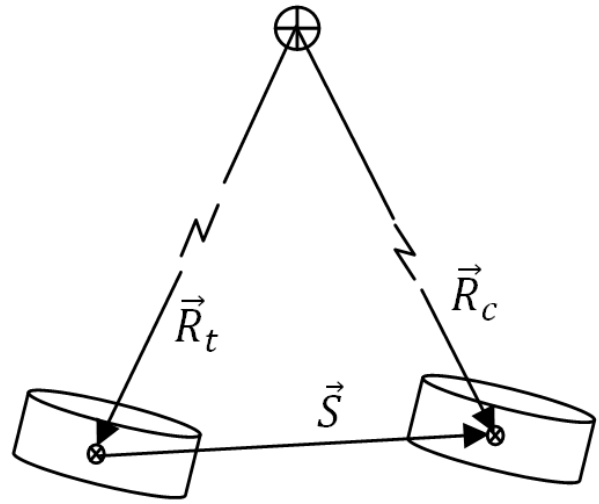

Fig. 1 Chaser (c) and target configuration in orbit.

The model shown in figure 1 is used to derive the translational equations. $\vec{R}_C$ and $\vec{R}_t$ are the inertial position vectors of the spacecraft chaser c and target t, respectively. The position vector $\vec{S}$ refers to the relative position between both spacecraft. The symbol $\otimes$ means the center o mass of the spacecraft. The symbol $\oplus$ means the Earth.

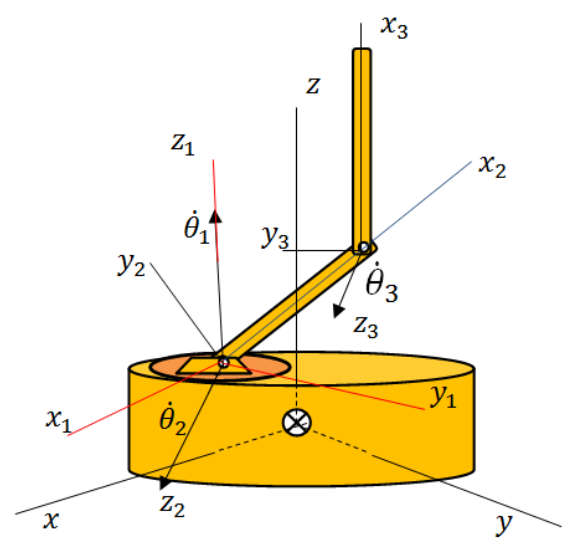

Fig. 2 Satellite like a robótic manipulator

Figure *2* shows the spacecraft like a robotic manipulator. The degrees-of-freedom associated with the manipulator dynamics is θ_i i =*1,2, 3.* The angular rates are represented by arrows. The frame standards were defined according to the Denavit-Hartenberg convention

as used in the Craig's classical book [2]. The rotation degree-of-freedom for each link is a z-axis. A rotation of an angle α about a x-axis is used to generate a new z-axis when needed. Those interested in the details of this convention can find it in [2]. The Lagrangian formulation for generalized coordinates [3] has been used to derive the manipulator equations while the Lagrangian formulation for quasi-coordinates [3] has been used to derive the modified Euler's Equations for the attitude motion. In the computer simulations the quaternion has been used to replace the kinematic equations to prevent singularity problems while simulating the attitude equations of motion via computer. A set of algebraic equations has been used to pass from quaternion to attitude angles (roll, pitch, and yaw in this work) when showing the results.
The modified Euler's Equations exhibits the several coupling for this problem, i.e., the crossing terms with the robotic and the orbital rates. The terminology "modified" has to do with those coupling terms.
For both Lagrange formulations, the approach is the same. It is necessary do obtain the energies (Kinetic and Potential (gravitational, elastic, electric if any). In the problem approached in this paper the potential energy is not considered. Instead the gravity-gradient torque is derived and included as external torque in the Euler's Equations. The Kinetic energy for the complete system can be obtained by summing the kinetic energies for the chaser platform, and the kinetic energy for each link of the robotic manipulator. The same approach can be used to obtain the equations of motion for the target spacecraft. The target is considered as a rigid spacecraft and the Euler's equations include only the orbital rate to make the equations compatible with the requirement that the Euler's equations are valid only in inertial system of reference.
The kinetic energy for the chaser is given by

$$T = T_p + \sum_{i-1}^{3} T_{\ell_i} \qquad (1)$$

$$T_p = \frac{1}{2} I_{px} {\omega_x}^2 + \frac{1}{2} I_{px} {\omega_y}^2 + \frac{1}{2} I_{px} {\omega_z}^2 \qquad (2)$$

$$T_{\ell_1} = \frac{1}{2}\Big[\big(I_{x_1} c^2\theta_1 + I_{y_1} s^2\theta_1\big)\omega_x^2 + \big(I_{x_1} s^2\theta_1 + Iy1c2\theta1\omega y2 + Iz1\omega za + \theta 12 \qquad (3)$$

$$\begin{aligned} T_{\ell_2} = {} & \frac{1}{2}\big(I_{y_2} s^2\theta_1 c^2\theta_2 + I_{z_2} s^2\theta_2\big)\big(\omega_x + s\theta_1\dot{\theta}_2\big)^2 \\ & + \frac{1}{2}\big(I_{y_2} s^2\theta_1 s^2\theta_2 \\ & + I_{z_2} c^2\theta_2\big)\big(\omega_y - c\theta_1\dot{\theta}_2\big)^2 \\ & + \frac{1}{2} I_{y_2} c^2\theta_1\big(\omega_{z_a} + \dot{\theta}_1\big)^2 \\ & + \big(I_{z_2} - I_{y_2} s^2\theta_1\big) s\theta_2 c\theta_2 \Big(\omega_x\omega_y \\ & - \omega_x c\theta_1\dot{\theta}_2 + \omega_y s\theta_1\dot{\theta}_2 - s\theta_1 c\theta_1 {\dot{\theta}_2}^2\Big) \\ & - I_{y_2} s\theta_1 c\theta_1 c\theta_2\big(\omega_x + s\theta_1\dot{\theta}_2\big)\big(\omega_{z_a} \\ & + \dot{\theta}_1\big) \\ & + I_{y_2} s\theta_1 c\theta_1 s\theta_2\big(\omega_y - c\theta_1\dot{\theta}_2\big)\big(\omega_{z_a} \\ & + \dot{\theta}_1\big) \end{aligned} \qquad (4)$$

$$\begin{aligned} T_{\ell_3} = {} & \frac{1}{2}\big[I_{y3} s^2\theta_1 c^2(\theta_2 + \theta_3) \\ & + I_{z3} s^2(\theta_2 + \theta_3)\big]\big[\omega_x \\ & + s\theta_1\big(\dot{\theta}_2 + \dot{\theta}_3\big)\big]^2 \\ & + \frac{1}{2}\big[I_{z3} c^2(\theta_2 + \theta_3) \\ & + I_{y3} s^2\theta_1 s^2(\theta_2 + \theta_3)\big]\big[\omega_y \\ & - c\theta_1\big(\dot{\theta}_2 + \dot{\theta}_3\big)\big]^2 \\ & + \frac{1}{2} I_{y3} c^2\theta_1\big(\omega_{z_a} + \dot{\theta}_1\big)^2 \\ & - \big[\big(I_{z3} - I_{y3} s^2\theta_1\big) s(\theta_2 + \theta_3) c(\theta_2 \\ & + \theta_3)\big]\big[\omega_x + s\theta_1\big(\dot{\theta}_2 + \dot{\theta}_3\big)\big]\big[\omega_y \\ & - c\theta_1\big(\dot{\theta}_2 + \dot{\theta}_3\big)\big] \\ & - I_{y3} s\theta_1 c\theta_1 c(\theta_2 + \theta_3)\big[\omega_x \\ & + s\theta_1\big(\dot{\theta}_2 + \dot{\theta}_3\big)\big]\big(\omega_{z_a} + \dot{\theta}_1\big) \\ & + I_{y3} s\theta_1 c\theta_1 s(\theta_2 + \theta_3)\big[\omega_y \\ & - c\theta_1\big(\dot{\theta}_2 + \dot{\theta}_3\big)\big]\big(\omega_{z_a} + \dot{\theta}_1\big) \end{aligned} \qquad (5)$$

It was defined the system of axes xyz as the principal axis spacecraft platform (*p*) is out of work and in its indicial condition of zero angles and zero rates. This assumption allows as writing the kinetic energy for the platform as

Where

I_{px}, I_{py}, and I_{pz} are the principal moments of inertia for the spacecraft platform;
ℓ_i $i = 1,2,3$ are links of the robotic manipulator

I_{x_i} $i=1,2,3$ are the moments of inertia of with respect to $x-axis$ for the ℓ_i $i=1,2,3;$

I_{y_i} $i=1,2,3$ are the moments of inertia of with respect to $y-axis$ for the ℓ_i $i=1,2,3;$

I_{z_i} $i=1,2,3$ are the moments of inertia of with respect to $z-axis$ for the ℓ_i $i=1,2,3;$

$\{\omega_x \quad \omega_y \quad \omega_z\}^T$ is angular velocity vector. The elements of the vector are the angular rates.

$s\theta_i$ and the $c\theta_i$ $i=1,2$ are the $sin\theta_i$ and $cos\theta_i$, respectively

$s(\theta_2+\theta_3)$ and $c(\theta_2+\theta_3)$ are the $sin(\theta_2+\theta_3)$ and $cos(\theta_2+\theta_3)$, respectively.

$\dot{\theta}_i$ $i=1,2,3$ are the rates of the robotic manipulator

$\omega_{z_a}=\omega_z+n$, where n is orbital rate.

The equations of motion are given by the Lagrange's formulas

$$\frac{d}{dt}\left\{\frac{\partial T}{\partial\omega}\right\}+\{[\widetilde{\omega}]\}\left\{\frac{\partial T}{\partial\omega}\right\}=\{Ng+Nm\}+\{Nc\} \qquad (6)$$

For quasi-coordinates, and

$$\frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\theta}_i}\right)-\frac{\partial T}{\partial\theta_i}=Q_i \qquad (7)$$

Where

$\{Ng+Nm\}$ is the perturbation due to the gravity-gradient and the residual magnetic field inside the satellite. $\{Nc\}$ is the control torque torque The kinematics is written in terms of quaternion [*4*] as

$$\begin{Bmatrix}\dot{q}_1\\ \dot{q}_2\\ \dot{q}_3\\ \dot{q}_4\end{Bmatrix}=\begin{bmatrix}0 & \Delta\omega_z & -\Delta\omega_y & \Delta\omega_x\\ -\Delta\omega_z & 0 & \Delta\omega_x & \Delta\omega_y\\ \Delta\omega_y & -\Delta\omega_x & 0 & \Delta\omega_z\\ -\Delta\omega_x & -\Delta\omega_y & -\Delta\omega_z & 0\end{bmatrix}\begin{Bmatrix}q_1\\ q_2\\ q_3\\ q_4\end{Bmatrix} \qquad (8)$$

Where
$\Delta\omega_\alpha=\omega_\alpha(\varphi,\theta,\psi,\dot{\varphi},\dot{\theta},\dot{\psi})-n\ell_z$; and

$$\ell_z=2\begin{Bmatrix}q_1q_3-q_2q_4\\ q_2q_3-q_1q_4\\ {q_3}^2+{q_4}^2-\frac{1}{2}\end{Bmatrix} \qquad (9)$$

The return to attitude angles are obtained by [*4*]

$$\varphi=\tan^{-1}\frac{q_1q_4+q_2q_3}{{q_3}^2+{q_4}^2-\frac{1}{2}} \qquad (10)$$

$$\theta=-\sin^{-1}[2(q_1q_3-q_2q_4)] \qquad (11)$$

$$\psi=\tan^{-1}\frac{q_1q_2+q_4q_3}{{q_1}^2+{q_4}^2-\frac{1}{2}} \qquad (12)$$

The resulting Euler's Equations obtained by is a complicated system of equations. $\left\{\frac{\partial T}{\partial\omega}\right\}$ yields in fact angular momentum of the system. It can be written as

$$[H]=[I_T]\{\omega\}$$

Where

$$[I_T]=[I_P]+\sum_{i=1}^{3}[I_{\ell_i}] \qquad (13)$$

So the term $\frac{d}{dt}\left\{\frac{\partial T}{\partial\omega}\right\}$ can be written as

$$[I_T]\{\dot{\omega}\}+[\dot{I}_t]\{\omega\}$$

which leads to a large expression in the time derivatives of inertia matrices of the links expressed in the satellite frame. These equations are not shown here. The same happen with the equations of motion for $\ddot{\theta}_i$ $i=1,2,3$ obtained by the Lagrange's formula for generalized coordinates.

The mathematical model for the target can be written as

$$[I_t]\{\dot{\omega}_t\}+[\widetilde{\omega}_t][I_t]\{\omega_t\}=\{N_t\} \qquad (14)$$

Where $[I_t]$ is target matrix of inertia and $[\widetilde{\omega}]$ is a skew symmetric matrix in the components of $\{\omega_t\}$

The relative attitude motion, between chaser and target spacecraft, denoted by q_r, is derived by using the quaternion product. The quaternion product between two quaternion vectors q_1 and q_2 and the rotation matrix that relates the spacecraft frame with to the inertial

frame. Details of the formulation are presented in [5]. Thought this formulation a relative rotation matrix can be written as

$$[{}^{c}_{t}R(q_r)] = [{}^{c}_{i}R(q_c)][{}^{t}_{i}R(q_c)] \quad (15)$$

Alternatively written as

$$[{}^{c}_{t}R(q_r)] = [{}^{c}_{i}R(q_c)^T][{}^{t}_{i}R(q_t)] \quad (16)$$

Therefore the relative attitude model q_r is obtained from q_c and q_t as

$$q_r = q_c \otimes {q_r}^{-1} \quad (17)$$

the relative angular velocity is obtained as follows

$$\{{}^{ct}_{ct}\omega\} = \{{}^{c}_{ic}\omega\} - \{{}^{c}_{it}\omega\} = [\{{}^{c}_{ic}\omega\} - [R(q_r)]\{{}^{t}_{it}\omega\}] \quad (18)$$

Defining $\{\omega_r\} = \{{}^{ct}_{ct}\omega\}$, $\{\omega_c\} = \{{}^{c}_{ic}\omega\}$, and $\{\omega_t\} = \{{}^{t}_{it}\omega\}$

By using these definitions the relative kinematics can be written as

$$\dot{q}_r = \frac{1}{2}[\Omega(\omega_r)]\{q_r\} \quad (19)$$

$$[\Omega(\omega_r)] = \begin{bmatrix} 0 & \{\omega_r\}^T \\ \{\omega_r\} & [\widetilde{\omega_r}] \end{bmatrix} \quad (20)$$

Once the mathematical model is obtained, we write the state equations. In the case of this paper the state is defined as

$$[x] = \{\theta_1\ \theta_2\ \theta_3\ \dot{\theta}_1\ \dot{\theta}_2\ \dot{\theta}_3\ \ q_1\, q_2\, q_3\ \ q_4\ \ \omega_x\, \omega_y\, \omega_{xz}\}^T = \{x_1\ \ x_2\ \ x_3\ \ x_4\ \ x_5\ \ x_6\ \ x_7\ \ x_8\ \ x_9\, x_{10}\ \ x_{11}\ \ x_{12}\ \ x_{13}\}^T \quad (21)$$

The linearized equations of motion for the translational motion can be written as [6] by taking the second time derivative of the chaser-target relative position $\vec{S} = \vec{R}_c - \vec{R}_t$, see figure 1. These associated equations of motion result from the application of the Newton's law of universal gravitation combined with the Newton's second law. The linear equations of motion requires the Taylor's series expansion for $\vec{S} = \vec{R}_c - \vec{R}_t$, as shown in [6].

$$\begin{aligned} f_X &= \ddot{x} - 2n\dot{z} \\ f_y &= \ddot{y} - 2n^2 y \\ f_z &= \ddot{z} - 2n\dot{x} - 3n^2 z \end{aligned} \quad (22)$$

Where f_x, f_y and f_z are the force components acting on the two spacecraft

The state vector is

$$\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{Bmatrix} = \begin{Bmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} \quad (23)$$

$$\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2n \\ 0 & -n^2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 3n^2 & -2n & 0 & 0 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 0 \\ u_x \\ u_y \\ u_z \end{Bmatrix} \quad (24)$$

Where u_x, u_y, and u_z are components of the control vector

Control Design

We have used the LQR technique for the relative translational equations. We included in the control in the system equations the linearized equations of the rotational motion without considering the robot manipulator just to synchronize the attitude between the target and the chaser spacecraft. The LQR technique can be state as

Minimize a cost function given by

$$J = \int_0^{\infty} \{x\}^T [Q]\{x\} + \{u\}^T [Q]\{u\}\, dt \quad (25)$$

Subject to

$$\{\dot{x}\} = [A]\{x\} + [B]\{u\} \quad (26)$$

$$\{u\} = -[k]\{x\} \quad (27)$$

$\{u\}$ is the state feedback control law that result of the quadratic performance index optimization subjected to the state equations. [*A*] and [*B*]are the plant and input matrices, respectively. [*Q*] is

a positive semi definite state weighting matrix, and R is the control weighting matrix that has to be positive definite. Q is $2n \times 2n$ and R is $p \times p$.

This technique requires the solution of the Riccati algebraic equation given by

$$[A]^T[P] + [P][A] + [Q] - [P][B][R]^{-1}[B]^T[P] = 0 \quad (28)$$

Where $[P]$ is the Riccati matrix.

The proportional Integral Derivate (PID) was used to control the nonlinear attitude equations for the chaser spacecraft. The proportional action provides a contribution which depends on the instantaneous value of the control error. The integral action, on the other hand, gives a controller output that is proportional to the accumulated error, which implies that it is a slow reaction control mode. And finally, the derivative action acts on the rate of change of the control error. The mathematical equation of a PID controller can be written as

$$N_c = k_p\theta_e + k_d\dot{\theta}_e + k_i \int \theta_e \, dt \quad (29)$$

Where θ_e is the pointing error and $k_p, k_d,$ and k_i are the PID controller gains.

PID gain selection

A simple gain selection strategy based on the Ziegler–Nichols [7] rule has been implemented. The proportional and derivative gains are obtained by taking into account the inertia of the system, the pulsation and the desired control damping. The gains are obtained by

$$k_p = -diag(I)\omega$$

$$k_d = -2diag(I)\omega\zeta \quad (30)$$

$$k_i = \frac{{k_p}^2}{2k_d}$$

where I, ω, and ζ are the inertia matrix, pulsation and damping, respectively.

Thruster control

The thrusters' control law is based on bang-bang control where the thruster characteristics are taken into account [7]. The thrusters control law is written as

$$N_t = Fd\frac{tOn}{dt} \quad (31)$$

Where $tOn = \frac{Ndt}{Fd}$, with *N* being the desired torque, *F* the thruster force, and *d* the distance from the center of mass. d*t* is the control sampling time.

Simulations and Results

The equations of motion have been integrated numerically by using the Matlab® software package. The results for the linear translational equations (relative motion) are presented in figures 4 to 6 and figure 8.The control technique is the LQR. The results show that the system behavior of the spacecraft in attitude and translation fulfill the objectives. The initial data for the sake of simulations, linear case, are

Inertia matrix

$$[I] = \begin{bmatrix} 80 & 0 & 0.003 \\ 0 & 100 & 0.001 \\ 0.003 & .001 & 120 \end{bmatrix}$$

Initial parameters:
Chaser
Yaw = 6 degrees
Pitch= 5 degrees
Roll= 0 degrees
Target
Yaw = 12 degrees
Pitch= 10 degrees
Roll= 0 degrees

Orbital rate =.0011 rd/s
Position: 5 m from the target
Relative rate: 7 mm/s
Angular rate: -5, 3, 3 degrees/s

The result of the simulations for this first case (linear equations) is shown in figures 3 to 8. The

main features to consider in this simulation are the relative motion in position and attitude and the control effort,

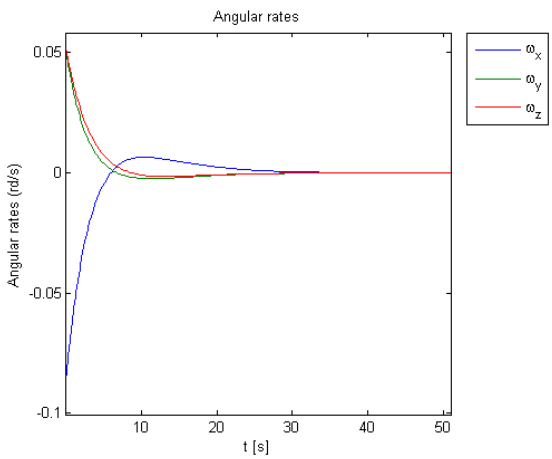

Fig. *3* Angular rates decay for the system under control

Figure *3* shows the behavior of the relative angular velocity under the control. Figure 4 shows the attitude synchronization between both spacecraft.
Figure *4* shows one important feature of the RVB/B, the attitude synchronization. The chaser spacecraft align its attitude with that of the target.

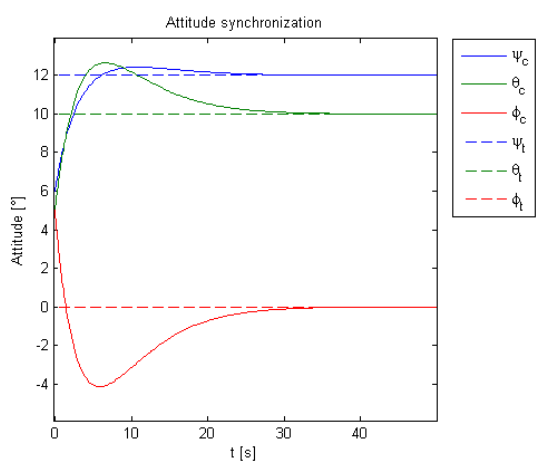

Fig *4* Attitude synchronization between both spacecraft

Figure *5* show the reduction of the relative distance between the chaser and the target. The berthing position must be planned carefully to avoid collision and to guarantee the work space of the manipulator.

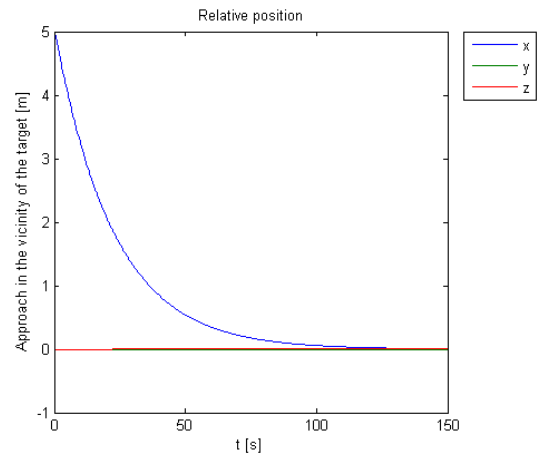

Fig *5* The relative position with respect to time

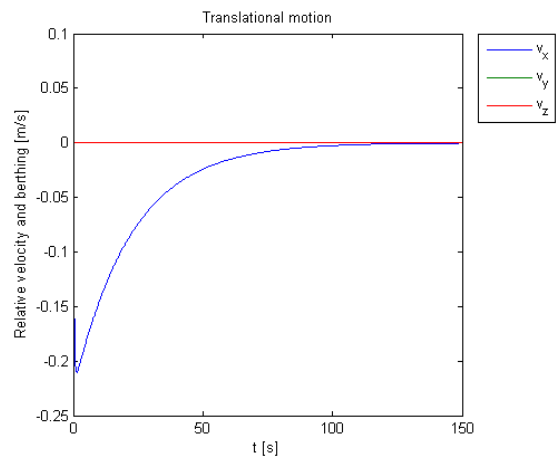

Fig *6* Relative velocity with respect to time

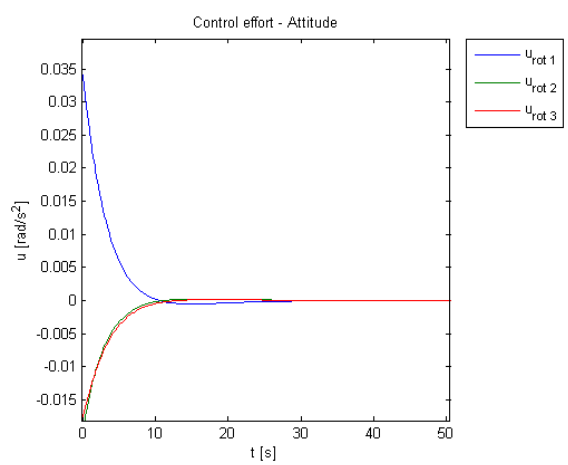

Fig *7* The control effort in to control the attitude.

Figure 6 shows the result of the control to reach the configuration of zero relative velocity. This is very important for berthing operations. It is a

condition in which both spacecraft are berthed with respect to each other.

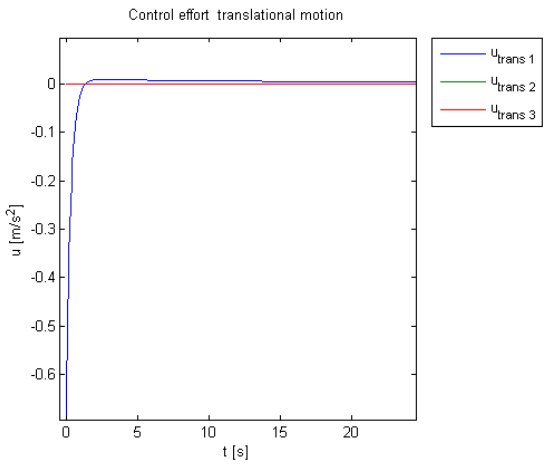

Fig. *8* Control efforts - translational motion

Figures *8* and *9* show the control effort for the attitude control and for the translational motion control, respectively. All this results is for the linear system of equations for both the translational and rotational motions.

The next computer simulations are for the nonlinear equations for the attitude motion. The control technique is the PID. The actuators are reaction wheels and thrusters. The initial inertia matrix is the same as the case for the linear system but it vary with time since it depends on the angles associated with robotic manipulator link motions. So the inertia matrix is not constant. The PID gains are computed at each step of integration (sampling time) assumed as 0.5 seconds in this work. The nominal reaction wheel torque capacity is $20\ Nm$.

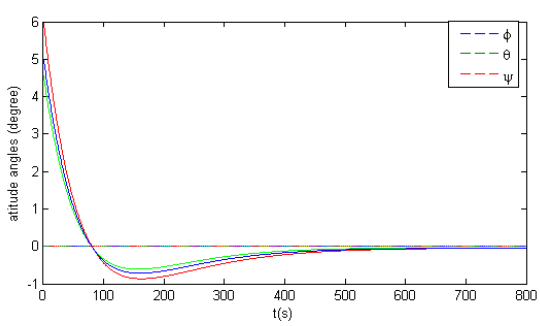

Fig. *9* Attitude control for the nonlinear system, using PDI technique.

Figure *9* shows the result for the configuration as follow: the robotic manipulator is not out of service and the reference is zero for roll, pitch and yaw.

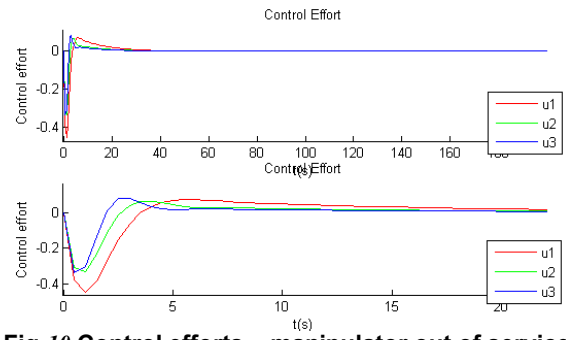

Fig *10* Control efforts – manipulator out of service, nonlinear system

Figure *10* shows the control effort when the robotic manipulator is out of service. By comparing this results with that of figure 9 it is clear the impact of the robotic manipulator dynamics on the control effort.

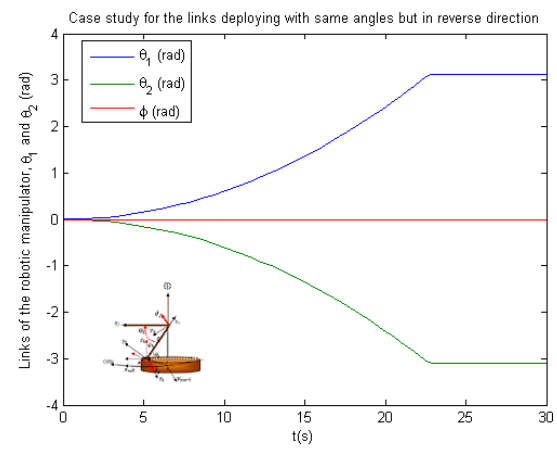

Fig *11* Case study for a robotic manipulator deploying the links in reverse sense and the same angles.

Figure *11* shows the result for a case study in which the links of the manipulator deploy in a synchronized way but with the angles in reverse sense. The results are as expected. The motions of the links in reverse sense cancel the reaction torque on the platform. The strategy to analyze this case was as follow: first the attitude control subsystem took attitude from a few degrees (as shown in figure *10*) to zero with the robotic manipulator out of service. Then the manipulator starts working by moving both links with the control subsystem off.

Conclusion

This paper presents and discusses the RVD/B operation under the point of view of control of attitude and translational motion with focus on the relative motion. The chaser-target relative motions approach is discussed including the relative attitude. A case study is considered for the synchronized angular motion of the links of the robotic manipulator deploying in reverse sense. The result was as expected in the sense that the reverse motion should cancel the robotic manipulator excitation on the platform attitude motion. The problem has been simulated via Matlab and the results show the performance of the PID and the LQR for the nonlinear and linear equations, respectively.

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